Security Monitoring & Wazuh SIEM Deployment

📖 Overview

This document details the deployment and configuration of **Wazuh SIEM (Security Information and Event Management)** as the cornerstone of the Blue Team security monitoring infrastructure. The implementation establishes comprehensive security monitoring capabilities using an all-in-one Wazuh deployment on **Rocky Linux 9.6**, strategically positioned within the dedicated **BlueTeam VLAN**.

🎯 Deployment Objectives

Primary Goals

- Deploy latest **Wazuh SIEM (version 4.12.0)** for comprehensive security monitoring
- Establish centralized log collection and analysis capabilities
- Create foundation for threat detection and incident response
- Implement security monitoring without impacting other network segments

Strategic Placement

- **VLAN Assignment**: BlueTeam VLAN (VLAN 20 - `192.168.20.0/24`)
- **Network Isolation**: Separated from other lab functions
- **Security Focus**: Dedicated environment for defensive security operations

🖥️ Server Preparation & Network Configuration

Hardware Specifications

Component	Details
Hostname	`lab-core-rocky`
Operating System	Rocky Linux 9.6 (Blue Onyx)
Architecture	x86_64
Network Interface	`enp0s20f0u4`
Switch Connection	Port 4 (Untagged VLAN 20)

Network Configuration Process

Initial Interface State Assessment

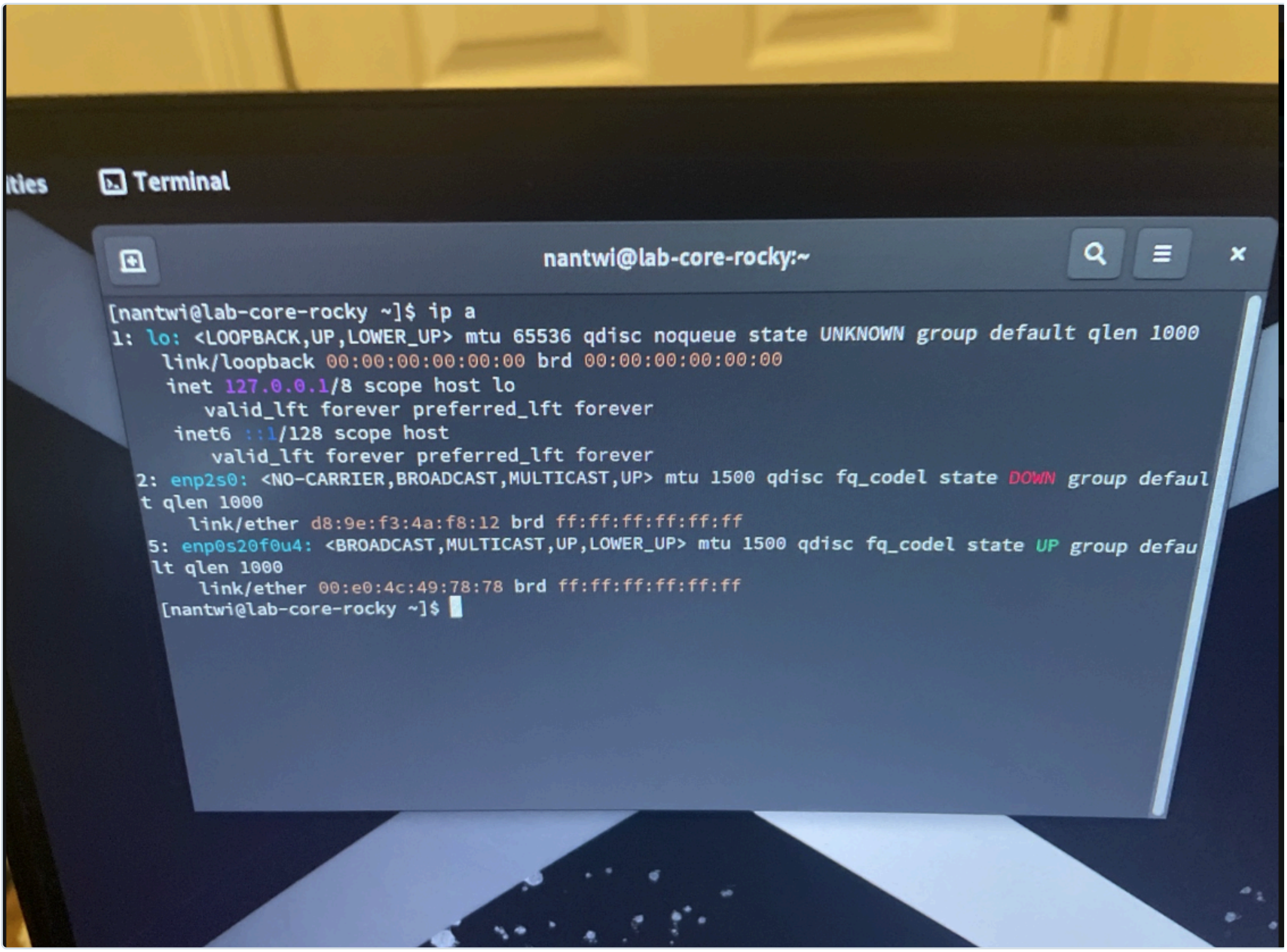
After connecting the server to **Port 4** and booting Rocky Linux:

Interface Status Check:

```
ip a
```

Observed State: - Interface `enp2s0`: **DOWN** (unused) - Interface `enp0s20f0u4`: **UP** (active, no IP assigned)

Network Setup Evidence



DHCP Verification & Testing

Before implementing static IP configuration, DHCP connectivity was verified to confirm proper VLAN assignment and network functionality.

DHCP Configuration Commands

```
# Create new network connection with DHCP
sudo nmcli con add type ethernet ifname enp0s20f0u4 con-name BlueTeam-DHCP ipv4.method auto

# Activate the connection
sudo nmcli con up BlueTeam-DHCP
```

DHCP Test Results

- ✅ **Dynamic IP Assignment**: `192.168.20.50/24` (within expected DHCP range)
- ✅ **Gateway Connectivity**: Successfully pinged `192.168.20.1`
- ✅ **VLAN Verification**: Confirmed placement in BlueTeam VLAN

Static IP Implementation

For production SIEM deployment, a **predictable static IP address** was required to ensure consistent access and monitoring capabilities.

Target Static Configuration

Parameter	Value
IP Address	`192.168.20.2/24`
Gateway	`192.168.20.1`
DNS Servers	`8.8.8.8`
Connection Name	`BlueTeam-Static`

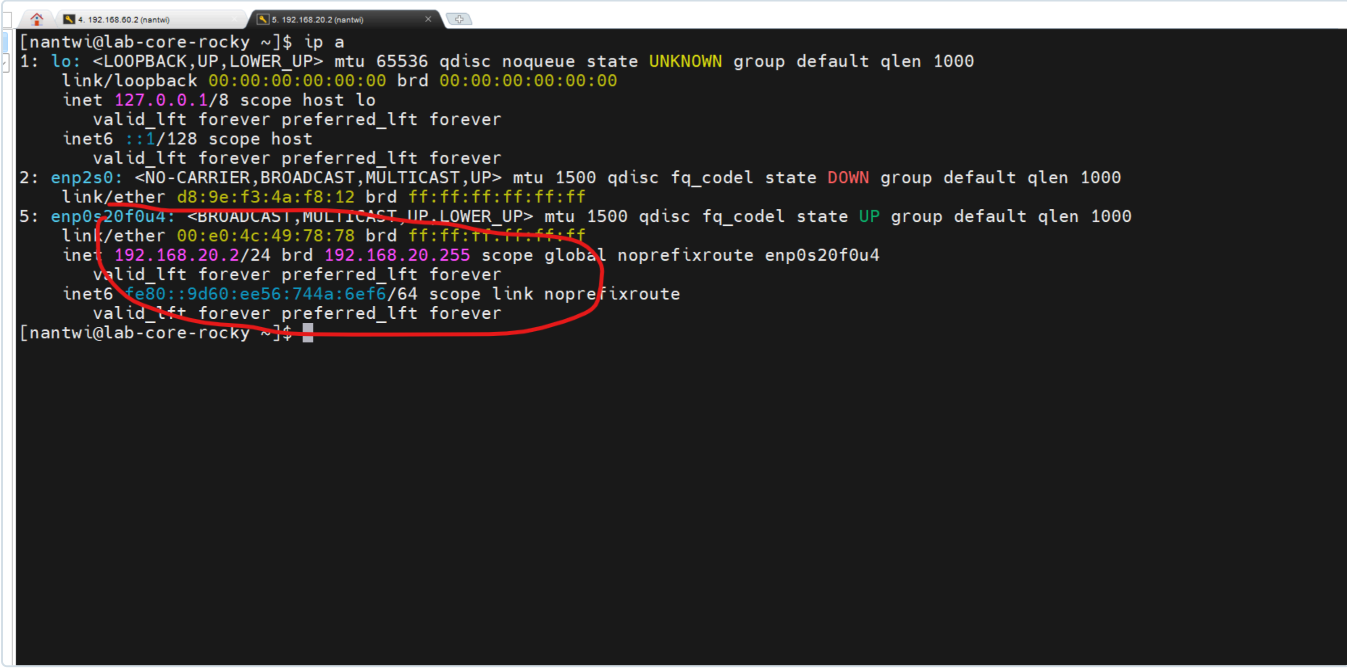
Static Configuration Commands

```
# Modify existing connection to static configuration
sudo nmcli con mod BlueTeam-DHCP \
  ipv4.method manual \
  ipv4.addresses 192.168.20.2/24 \
  ipv4.gateway 192.168.20.1 \
  ipv4.dns 8.8.8.8

# Restart connection to apply changes
sudo nmcli con down BlueTeam-DHCP
sudo nmcli con up BlueTeam-DHCP

# Rename connection for clarity
sudo nmcli con mod BlueTeam-DHCP connection.id BlueTeam-Static
```

Configuration Verification



Connectivity Validation

```
# Verify static IP assignment
ip a

# Test gateway connectivity
ping -c 4 192.168.20.1

# Verify internet connectivity
ping -c 4 8.8.8.8
```

Results: ✅ All connectivity tests successful - static configuration operational

📦 Wazuh SIEM Deployment Process

Pre-Installation System Preparation

System Updates & Dependencies

```
# Update system packages
sudo dnf update -y

# Install required utilities
sudo dnf install -y curl wget unzip gnupg2
```

Clean Previous Installations

Since Wazuh had been previously attempted on this system, comprehensive cleanup was performed to avoid conflicts:

```
# Remove any existing Wazuh packages
sudo dnf remove -y wazuh-* filebeat opensearch-dashboard

# Clean configuration and data directories
sudo rm -rf /etc/wazuh-* /var/lib/wazuh-* /usr/share/wazuh-* /etc/filebeat /var/lib/filebeat
```

Wazuh Installation Implementation

Installer Acquisition

```
# Download official Wazuh all-in-one installer
curl -sO https://packages.wazuh.com/4.12/wazuh-install.sh

# Make installer executable
chmod +x wazuh-install.sh
```

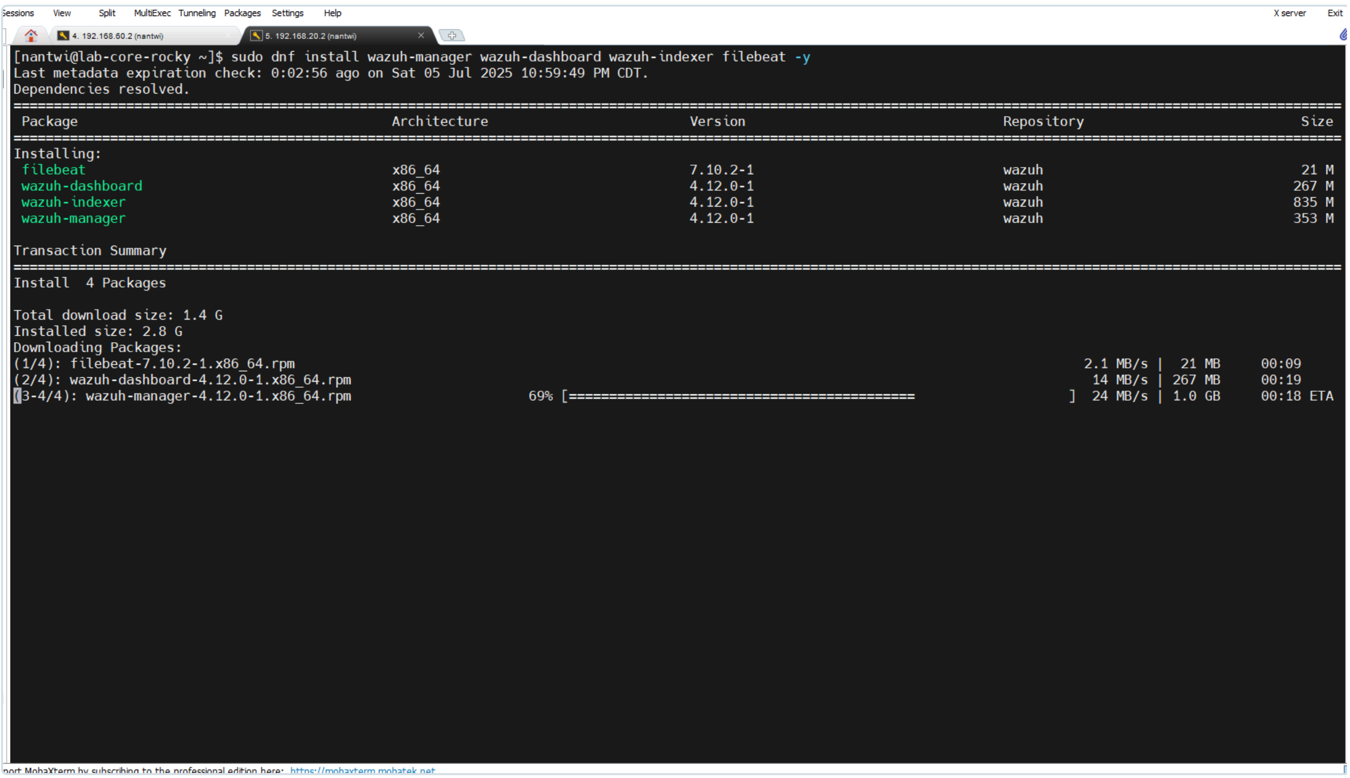
All-in-One Deployment

The Wazuh team's recommended deployment method was used for comprehensive SIEM functionality:

```
# Execute all-in-one installation
sudo ./wazuh-install.sh -a -i -o
```

Installation Parameters: - `-a` : Install all components (Manager, Indexer, Dashboard, Filebeat) - `-i` : Ignore OS compatibility check (Rocky Linux support) - `-o` : Remove old installations if present

Installation Process



Installation Duration: Approximately 10-15 minutes
Components Installed: - **Wazuh Manager**: Core SIEM engine and rule processing - **Wazuh Indexer**: OpenSearch-based data storage and indexing - **Wazuh Dashboard**: Web-based management interface - **Filebeat**: Log shipping and data forwarding

Installation Completion

Upon successful installation, the system provided: - **Dashboard URL**: `https://192.168.20.2:443` - **Admin Credentials**: Automatically generated secure password - **SSL Certificates**: Automatically configured for HTTPS access

🔧 Post-Installation Configuration

Firewall Configuration

Rocky Linux ships with `firewalld` enabled by default. Essential Wazuh ports were opened for proper operation:

```
# Open Wazuh Manager ports
sudo firewall-cmd --permanent --add-port=1514/tcp  # Agent communication
sudo firewall-cmd --permanent --add-port=1514/udp  # Agent communication
sudo firewall-cmd --permanent --add-port=1515/tcp  # Agent enrollment

# Open Dashboard port
sudo firewall-cmd --permanent --add-port=443/tcp   # HTTPS web interface

# Apply firewall changes
sudo firewall-cmd --reload
```

Service Validation

Verification of all Wazuh components:

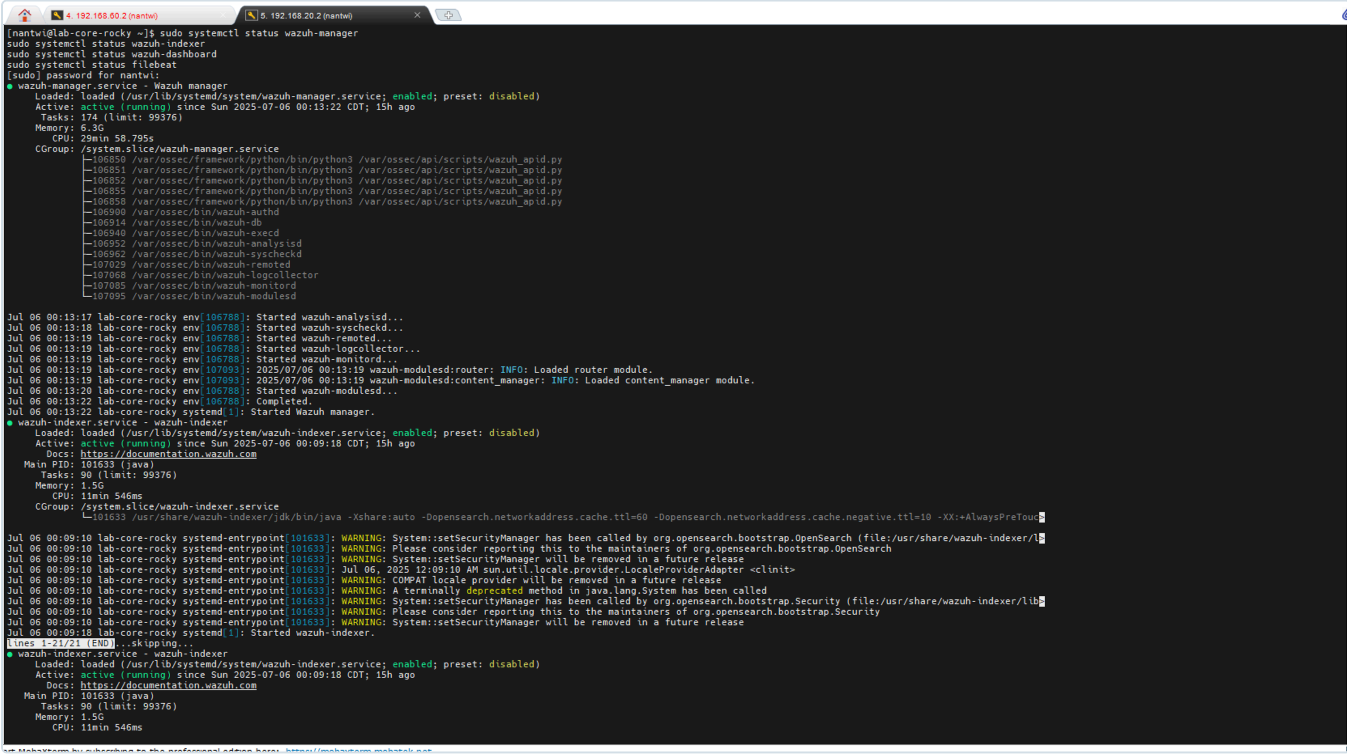
```
# Check Wazuh Manager status
sudo systemctl status wazuh-manager

# Check Wazuh Indexer status
sudo systemctl status wazuh-indexer

# Check Wazuh Dashboard status
sudo systemctl status wazuh-dashboard

# Check Filebeat status
sudo systemctl status filebeat
```

Service Status Verification



Service Status: ✅ All services active and running properly

📊 Wazuh Dashboard Access & Verification

SSH Access to Wazuh Server

Remote access to the Wazuh server instance for administration and troubleshooting:

SSH Connection Details:

```
# SSH into Wazuh server
ssh nantwi@192.168.20.2
```

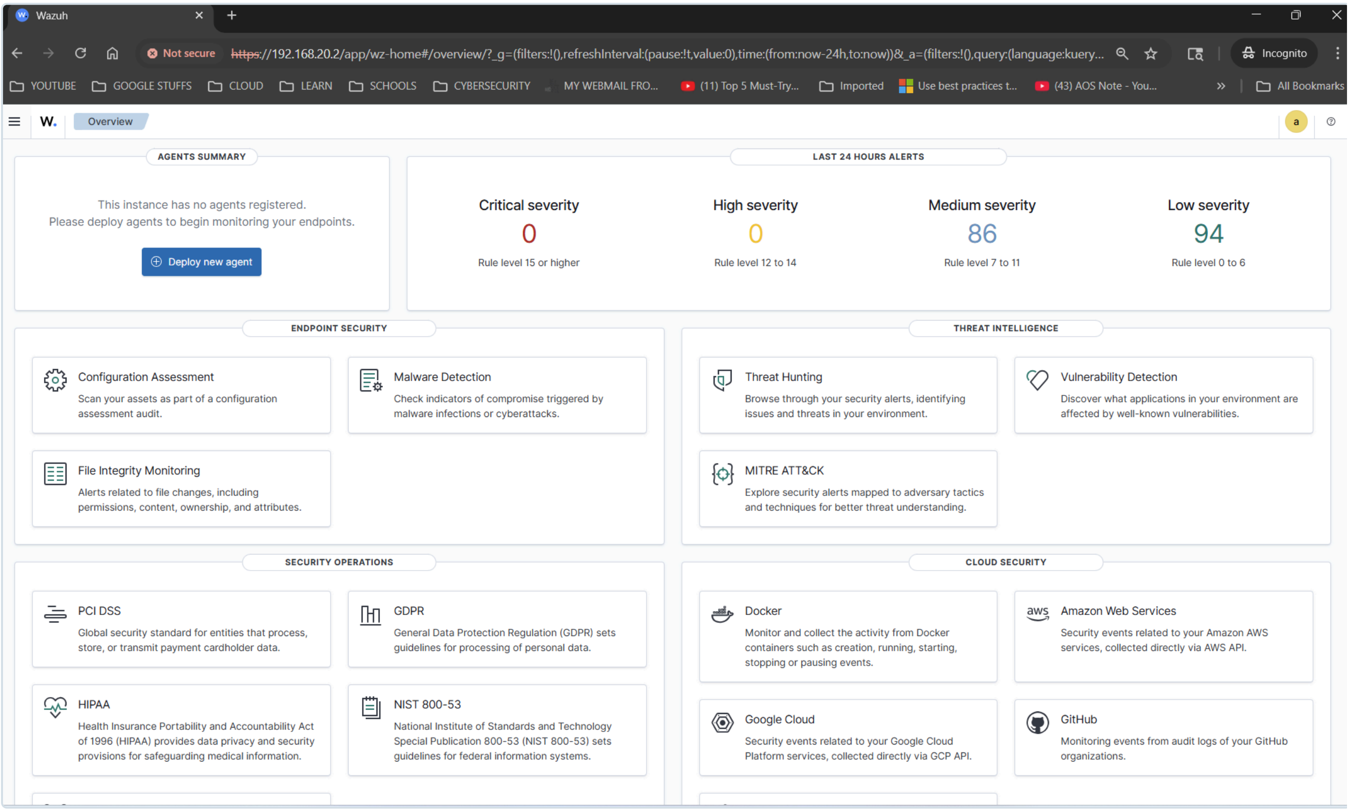
Parameter	Value
Hostname	`lab-core-rocky`
IP Address	`192.168.20.2`
SSH Port	`22` (default)
SSH Username	`nantwi`
VLAN	BlueTeam (VLAN 20)

Dashboard Accessibility

The Wazuh dashboard became accessible immediately after installation:

Access URL: `https://192.168.20.2`
Protocol: HTTPS with automatically generated SSL certificates
Authentication: Admin credentials provided during installation

Dashboard Interface



Wazuh Version Confirmation

- **Wazuh Version**: 4.12.0 (Latest stable release at deployment time)
- **Deployment Type**: All-in-One configuration
- **Components**: Manager, Indexer, Dashboard, Filebeat all operational

Initial Dashboard Features

- **Security Events**: Real-time security event monitoring
- **Agent Management**: Ready for endpoint agent deployment
- **Rule Management**: Pre-configured security rules and decoders
- **Compliance Dashboards**: Built-in compliance reporting
- **Threat Intelligence**: Integrated threat detection capabilities

🔐 Security Implementation

Network Security

- **VLAN Isolation**: SIEM contained within BlueTeam VLAN (192.168.20.0/24)
- **Firewall Protection**: pfSense controls all inter-VLAN communication

- **SSL Encryption**: Automatic HTTPS certificate generation and implementation
- **Access Control**: Dashboard access restricted to authenticated users
- **Port Security**: Only necessary ports opened in firewall configuration

Authentication & Access

- **Default Credentials**: Secure admin password automatically generated
- **HTTPS Enforcement**: All dashboard communication encrypted
- **Certificate Management**: SSL certificates automatically provisioned
- **Session Security**: Web-based authentication with session management

🎯 Operational Capabilities

Current SIEM Functionality

✅ **Real-time Monitoring**: Active security event collection and analysis
✅ **Log Management**: Centralized log aggregation and storage
✅ **Rule Engine**: Pre-configured security detection rules operational
✅ **Dashboard Interface**: Web-based management and monitoring
✅ **Agent Framework**: Ready for endpoint agent deployment

Built-in Security Features

- **Intrusion Detection**: File integrity monitoring capabilities
- **Vulnerability Assessment**: Integrated vulnerability scanning
- **Compliance Reporting**: PCI DSS, GDPR, HIPAA compliance frameworks
- **Threat Intelligence**: Integration with external threat feeds
- **Incident Response**: Automated response and alerting capabilities

🚀 Integration with Lab Infrastructure

Network Integration

- **VLAN Placement**: Strategic positioning in BlueTeam VLAN
- **Gateway Access**: Proper routing through pfSense (192.168.20.1)
- **Internet Connectivity**: Outbound access for threat intelligence updates
- **Inter-VLAN Security**: Controlled access from Management VLAN for administration

Administrative Access

- **Management VLAN Access**: pfSense rules allow administrative connectivity
- **SSH Access**: Available for system administration and maintenance
- **Web Dashboard**: HTTPS access for security operations
- **Remote Monitoring**: Integration prepared for Tailscale mesh VPN

📈 Performance & Scalability

System Resources

- **CPU Utilization**: Optimized for all-in-one deployment
- **Memory Usage**: Sufficient resources allocated for SIEM operations
- **Storage Capacity**: Adequate space for log retention and indexing
- **Network Performance**: Full gigabit connectivity through managed switch

Scalability Considerations

- **Agent Capacity**: Supports hundreds of endpoint agents
- **Log Volume**: Configurable retention policies for storage management
- **Performance Tuning**: Options for optimizing based on log volume
- **High Availability**: Foundation prepared for future HA implementation

🔄 Deployment Validation

Functional Testing

✅ **Service Status**: All Wazuh components operational
✅ **Web Interface**: Dashboard accessible and responsive
✅ **Network Connectivity**: Full connectivity to gateway and internet
✅ **Firewall Rules**: Proper port access configured
✅ **SSL Certificates**: HTTPS encryption functional

Security Verification

✅ **VLAN Isolation**: Confirmed placement in BlueTeam VLAN
✅ **Access Controls**: Administrative access properly restricted
✅ **Service Security**: All services running with appropriate permissions
✅ **Network Security**: pfSense firewall rules effective

⚠️ Implementation Notes

Rocky Linux Compatibility

- **Official Support**: Rocky Linux not officially listed by Wazuh
- **Practical Reality**: Installation and operation fully stable
- **Compatibility Flag**: `-i` flag used to bypass OS check
- **Production Viability**: No functional limitations observed

Installation Method

- **All-in-One Choice**: Selected for comprehensive functionality in single system

- **Alternative Options**: Distributed deployment available for larger environments
- **Resource Efficiency**: Single-server deployment optimal for lab environment
- **Future Migration**: Can expand to distributed architecture if needed

🛠️ Maintenance & Operations

Regular Maintenance Tasks

- **System Updates**: Regular Rocky Linux security updates
- **Wazuh Updates**: Periodic SIEM platform updates
- **Rule Tuning**: Optimization of detection rules based on environment
- **Log Management**: Monitoring of storage usage and retention policies

Monitoring & Alerting

- **Service Health**: Monitor all Wazuh component status
- **Resource Usage**: Track CPU, memory, and storage utilization
- **Network Connectivity**: Ensure consistent network access
- **Security Events**: Regular review of security alerts and incidents

🎯 Next Phase Integration

Agent Deployment Planning

The SIEM infrastructure is now ready for: - **Linux Agent Deployment**: Ubuntu servers and workstations - **Windows Agent Integration**: Windows systems across lab environment - **Network Device Monitoring**: Integration with pfSense and switch logs - **Application Monitoring**: Custom log sources and application security

Security Operations Expansion

Prepared foundation for: - **Custom Rule Development**: Environment-specific detection rules - **Threat Hunting**: Proactive security investigation capabilities - **Incident Response**: Automated response and notification systems - **Compliance Reporting**: Regulatory compliance monitoring and reporting

✅ Deployment Summary

Successfully Implemented

- **Wazuh SIEM 4.12.0**: Latest version deployed and operational
- **All-in-One Configuration**: Complete SIEM stack on single system
- **BlueTeam VLAN Integration**: Properly segmented and secured
- **Network Connectivity**: Full access with appropriate security controls
- **Web Dashboard**: HTTPS-secured administrative interface

Current Operational Status

Component	Status	Accessibility
Wazuh Manager	🟢 Active	Internal network
Wazuh Indexer	🟢 Active	Internal components
Wazuh Dashboard	🟢 Active	`https://192.168.20.2`
Filebeat	🟢 Active	Log shipping operational

Infrastructure Integration

- **Network Foundation**: Successfully integrated with pfSense VLAN architecture
- **Security Posture**: Enhanced with centralized security monitoring
- **Operational Readiness**: Prepared for agent deployment and rule customization
- **Administrative Access**: Secured administrative interface and SSH access

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Security Monitoring Status: ✅ Complete and Operational
Next Phase: Observability Stack Deployment